

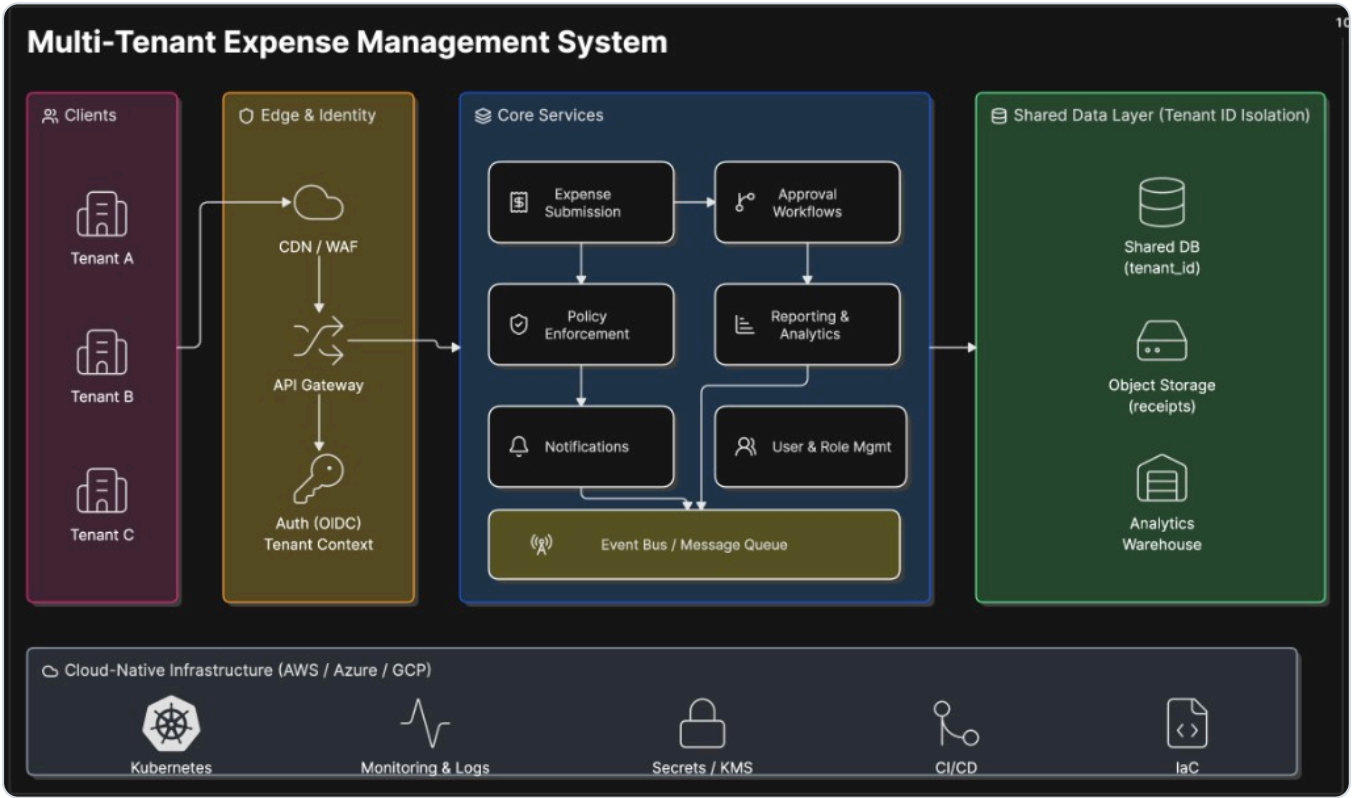
EMS — Technical Documentation (API Reference & UI Walkthrough)

This document describes the implemented Expense Management System: every API endpoint with request/response samples, with **UI screenshots shown inline next to the flow they belong to**.

- **Base URL:** `http://localhost:4000`
- **Format:** JSON over HTTP
- **Source design:** [level-2-design.md](#) · scope: [level-1.5-scope.md](#)

Architecture

Target / production architecture — multi-tenant clients behind an edge + API gateway, stateless core services on a shared data layer with `tenant_id` isolation, on cloud-native infra:



The as-built prototype implements the same core capabilities as a **modular monolith** (single Express app + PostgreSQL + optional Redis); the gateway, event bus, object storage, analytics warehouse and infra layers are the documented production path.

1. Conventions

Authentication

- Obtain a token via `POST /auth/login` or `POST /auth/signup` .
- Send it on every protected request: `Authorization: Bearer <accessToken>` .
- The token encodes `{ id, org_id, role, email }` . The server derives `org_id` from the token — clients never pass it. This is the tenant-isolation guardrail.

Roles & permissions

Permission	employee	manager	finance	admin
expense:create / read own	✓	✓	✓	✓
expense:read reportees	—	✓	—	✓
expense:read all	—	—	✓	✓
expense:approve	—	✓	✓	✓
policy:manage / budget:manage	—	—	✓	✓
user:manage	—	—	—	✓
analytics:view	—	✓	✓	✓

Headers

Header	When	Purpose
<code>Authorization: Bearer <token></code>	all protected routes	authentication
<code>Content-Type: application/json</code>	requests with a body	parsing
<code>Idempotency-Key: <unique></code>	approve/reject (optional)	prevents double-apply on retry

Error envelope

All errors share one shape:

```
{ "error": { "code": "FORBIDDEN", "message": "You do not have permission...", "details": null } }
```

HTTP	code	Meaning
400	BAD_REQUEST	Malformed/invalid payload
401	UNAUTHORIZED	Missing/invalid token
403	FORBIDDEN	Authenticated but not allowed
404	NOT_FOUND	Resource not in your tenant
409	CONFLICT	Invalid state transition / duplicate
422	UNPROCESSABLE	Valid shape but business rule failed (budget, currency, no policy)

Notes on data types

- Postgres `NUMERIC` columns are returned as **strings** (e.g. `"3000.00"`), preserving precision. Clients should parse as needed.
- Pagination: list endpoints accept `?limit=` (max 100) and `?offset=` .

2. Auth & Onboarding

POST /auth/signup — create a new organization (tenant onboarding)

Atomically creates an organization, its first **admin** user, a default approval policy and an org budget, then returns tokens (auto-login). Public.

Request

```
{
  "orgName": "Initech",
  "baseCurrency": "INR",
  "adminName": "Bill Lumbergh",
  "email": "admin@initech.test",
  "password": "password123"
}
```

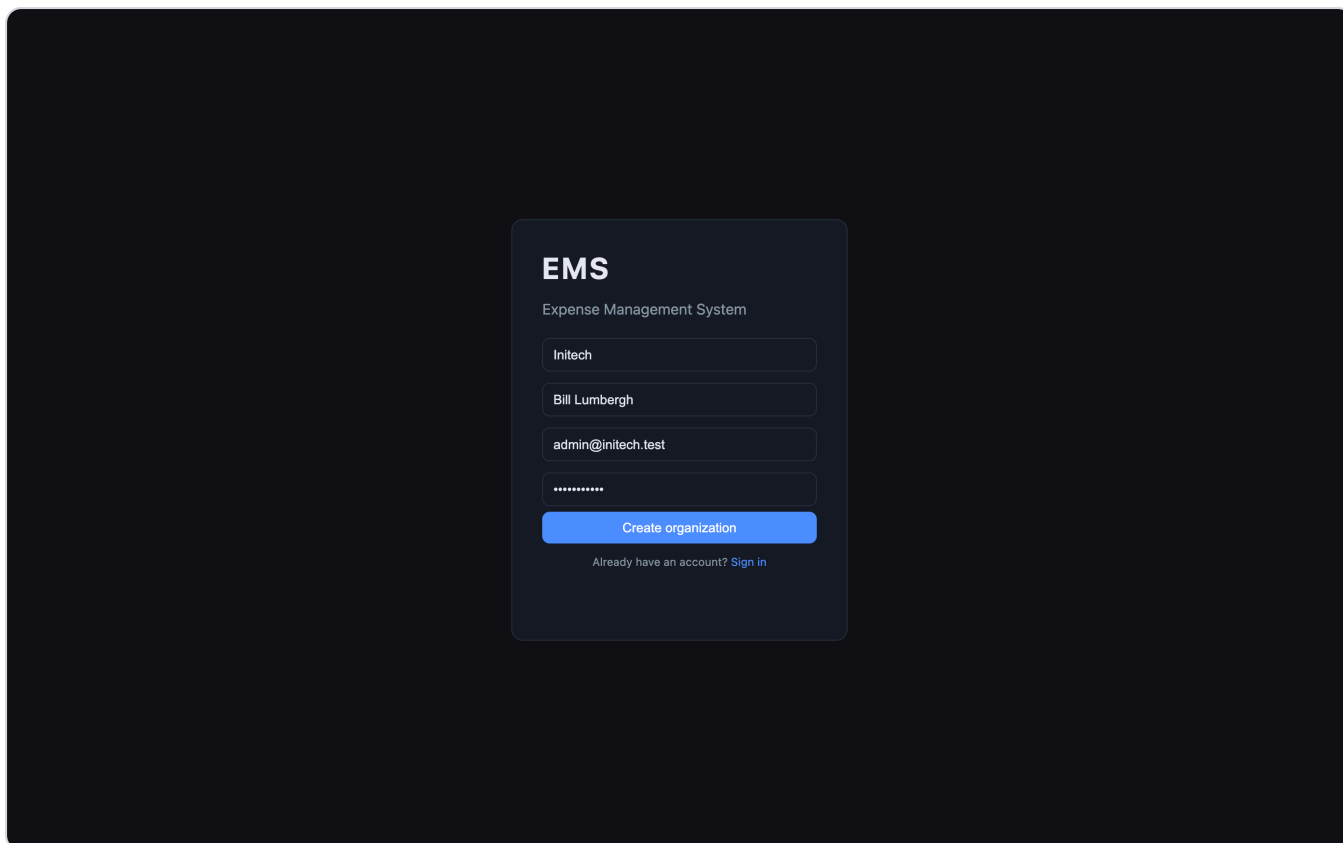
Field	Type	Required	Notes
orgName	string	yes	organization name
baseCurrency	string(3)	no	defaults <code>INR</code>
adminName	string	yes	first admin's name
email	string	yes	unique across all tenants
password	string	yes	min 6 chars

Response 201

```
{
  "accessToken": "eyJhbGciOi...",
  "refreshToken": "eyJhbGciOi...",
  "user": { "id": "uuid", "org_id": "uuid", "name": "Bill Lumbergh", "email": "admin@initech.test",
  "organization": { "id": "uuid", "name": "Initech", "base_currency": "INR" }
}
```

Errors: `400` invalid payload · `409` email already exists.

UI — Sign up (new organization): "Create an organization" provisions a new tenant + its first admin in one step, with a default policy so it's usable immediately.



POST /auth/login

Request

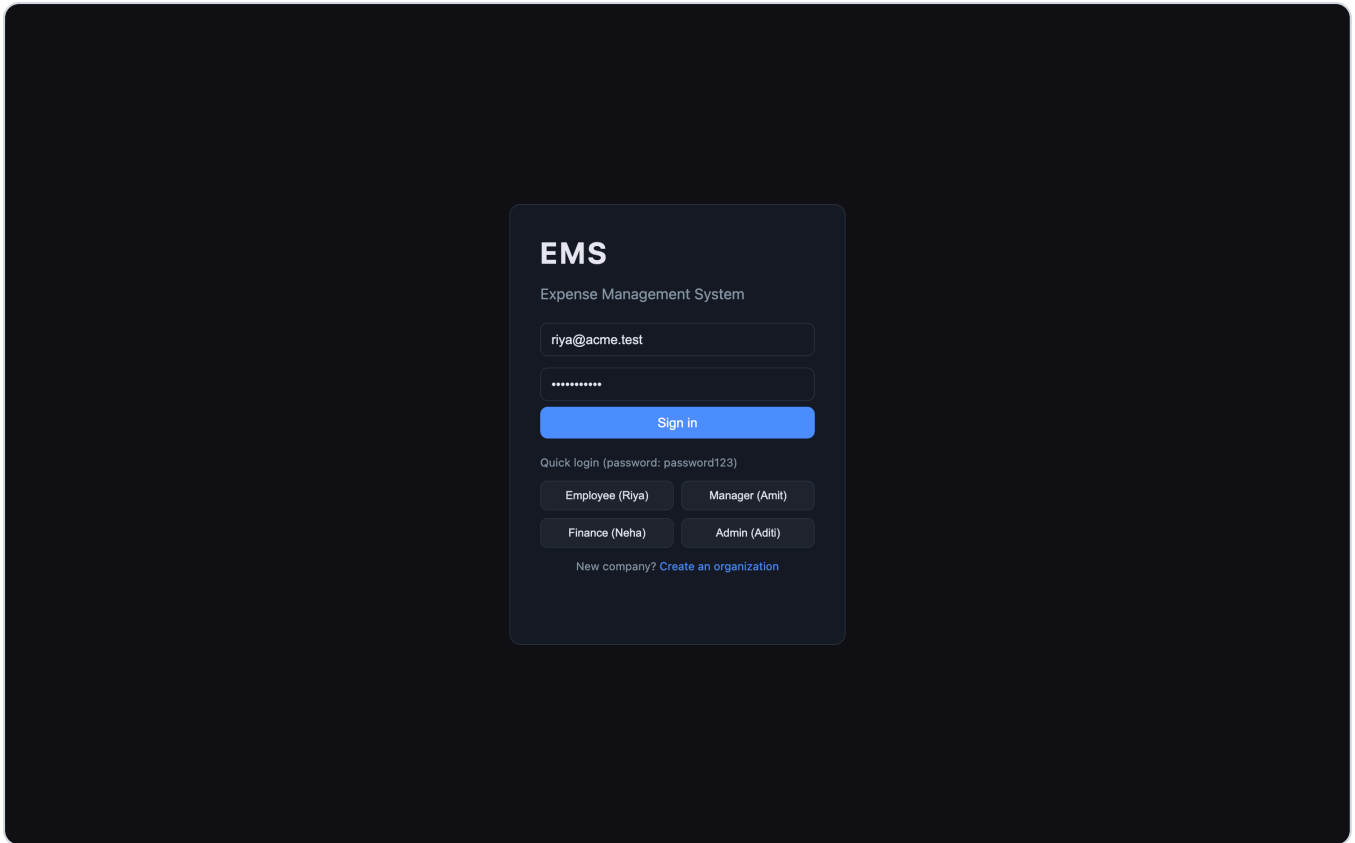
```
{ "email": "riya@acme.test", "password": "password123" }
```

Response 200

```
{
  "accessToken": "eyJhbGciOiJI...",
  "refreshToken": "eyJhbGciOiJI...",
  "user": {
    "id": "uuid", "org_id": "uuid", "name": "Riya Sharma",
    "email": "riya@acme.test", "role": "employee",
    "manager_id": "uuid", "is_active": true, "created_at": "2026-..."
  }
}
```

Errors: 401 invalid email or password.

UI — Login: sign in with email/password, or one-click **Quick login** for the seeded roles.



POST /auth/refresh — rotate refresh token

Validates an opaque refresh token, **revokes it (single-use rotation)** and issues a fresh access + refresh pair. Replaying a revoked token is treated as reuse and revokes all of the user's sessions.

Request

```
{ "refreshToken": "<opaque token from login/signup>" }
```

Response 200

```
{ "accessToken": "eyJhbGciOi...", "refreshToken": "<new opaque token>", "user": { "id": "uuid", "...":
```

Errors: 401 missing / invalid / expired / reused token.

POST /auth/logout

Revokes a refresh token. Idempotent. **Request:** { "refreshToken": "<token>" } · **Response 200** → { "revoked": true }

3. System

GET /health

Response 200 → { "status": "ok" }

GET /metrics

Prometheus text exposition. Public.

```
ems_requests_total 128
ems_errors_total 0
ems_request_latency_ms{quantile="0.5"} 4
ems_request_latency_ms{quantile="0.95"} 21
ems_request_latency_ms{quantile="0.99"} 39
```

4. Expenses

POST /expenses — create a draft

Permission: `expense:create` .

Request

```
{ "type": "reimbursement", "category": "travel", "description": "Client visit", "amount": 22000, "cu
```

Field	Type	Required	Notes
type	enum	yes	<code>reimbursement</code> <code>company_paid</code>
category	string	no	defaults <code>general</code>
description	string	no	
amount	number	yes	> 0, in <code>currency</code>
currency	string	no	defaults <code>INR</code> ; supported: INR, USD, EUR, GBP, AED, SGD

Response 201

```
{
  "id": "uuid", "org_id": "uuid", "requester_id": "uuid",
  "type": "reimbursement", "category": "travel", "description": "Client visit",
  "amount": "22000.00", "currency": "INR", "base_amount": "22000.00", "fx_rate": "1.000000",
  "status": "draft", "policy_snapshot_json": null, "current_level": 0,
  "sla_due_at": null, "created_at": "2026-...", "updated_at": "2026-..."
}
```

Errors: 400 invalid · 422 unsupported currency.

GET /expenses — list

Query: `scope=mine|reportees|all` (default `mine`), `status`, `type`, `limit`, `offset` .

- `reportees` requires `expense:read:reportees` ; `all` requires `expense:read:all` .

Response 200 → array of expense objects (newest first). **Errors:** 403 if scope exceeds your permission.

UI — Employee "My Expenses": create reimbursement or company-paid requests and track them with status pills. Foreign-currency amounts show the base (₹) conversion inline.

EMS

My Expenses

Riya Sharma · employee 2

Logout

Create expense

Reimbursement (self-paid)

travel

3000

INR

Business expense

Create draft

My expenses

TYPE	CATEGORY	AMOUNT	STATUS	
reimbursement	meals	₹4,500	in review	View
reimbursement	travel	₹16,600 (200.00 USD)	draft	View
reimbursement	travel	₹22,000	approved	View
reimbursement	meals	₹3,200	approved	View

GET /expenses/:id — get one (with approval chain)

Response 200

```
{
  "id": "uuid", "status": "in_review", "current_level": 1, "amount": "22000.00",
  "base_amount": "22000.00", "type": "reimbursement", "category": "travel",
  "steps": [
    { "id": "uuid", "level": 1, "required_role": "manager", "approver_id": "uuid", "status": "approved", "description": "..." },
    { "id": "uuid", "level": 2, "required_role": "finance", "approver_id": "uuid", "status": "pending", "description": "..." }
  ]
}
```

Errors: 404 not found / not in your tenant · 403 not allowed to view.

PATCH /expenses/:id — edit

Only the requester, only while `draft` / `submitted` / `in_review`. If the amount crosses a policy threshold while `in_review`, the approval chain is **rebuilt**. **Request** (any subset): `{ "amount": 20000, "currency": "INR", "category": "...", "description": "..." }` **Response 200** updated expense. **Errors:** 403, 409, 422.

DELETE /expenses/:id

Requester only, drafts only. **Response 200** → `{ "deleted": true }`. **Errors:** 403, 409.

POST /expenses/:id/submit

Moves `draft` → `in_review`, snapshots the policy, builds `ApprovalStep` s, notifies the first approver. **Response 200** → expense with `status:"in_review"`, `current_level:1`. **Errors:** 403 not requester · 409 not a draft · 422 budget exceeded / no active policy.

Budget-exceeded body:

```
{ "error": { "code": "UNPROCESSABLE", "message": "Budget exceeded for monthly (limit 2000000, already spent 1990000)", "details": { "ok": false, "period": "monthly", "limit": 2000000, "spent": 1990000 } } }
```

POST /expenses/:id/approve

Permission: `expense:approve`. Must be the assigned approver for the current level; cannot approve your own. Honors Idempotency-Key. **Request:** `{ "reason": "looks good" }` (optional). **Response 200** → expense; becomes `approved` after the last level, else stays `in_review` with `current_level` advanced. **Errors:** 403 wrong approver / self-approval · 404 · 409 not awaiting approval.

POST /expenses/:id/reject

Permission: `expense:approve`. **Reason is mandatory.** Terminates the chain. **Request:** `{ "reason": "not a business expense" }` **Response 200** → expense `status:"rejected"`. **Errors:** 400 no reason · 403 · 404 · 409.

POST /expenses/:id/withdraw

Requester cancels (`draft` / `submitted` / `in_review`). **Response 200** → `status:"withdrawn"`.

GET /expenses/:id/history — audit trail

Response 200

```
[
  { "id":"uuid","actor_id":"uuid","action":"expense.created","entity_type":"expense","entity_id":"uuid",
    { "action":"expense.submitted", "after_json":{"status":"in_review","levels":["manager","finance"]}},
    { "action":"expense.step_approved", "reason":"ok", "created_at":"2026-..." },
    { "action":"expense.approved", "created_at":"2026-..." }
  ]
```

UI — Expense detail (chain + audit trail): each expense shows its computed multi-level **approval chain** (level, role, status) and the immutable **history (audit trail)**. Requester actions (Submit/Withdraw) appear contextually.

EMS

My Expenses

Riya Sharma · employee 2Logout

Create expense

Reimbursement (self-paid)

travel

3000

INR

Business expense

Create draft

My expenses

TYPE	CATEGORY	AMOUNT	STATUS	
reimbursement	meals	₹4,500	in review	<button>View</button>
reimbursement	travel	₹16,600 (200.00 USD)	draft	<button>View</button>
reimbursement	travel	₹22,000	approved	<button>View</button>
reimbursement	meals	₹3,200	approved	<button>View</button>

Expense detail

Status

in review

Type

reimbursement

Amount

₹4,500 (4500.00 INR)

Category

meals

Current level

1

Approval chain

L1 · manager · pending

Withdraw

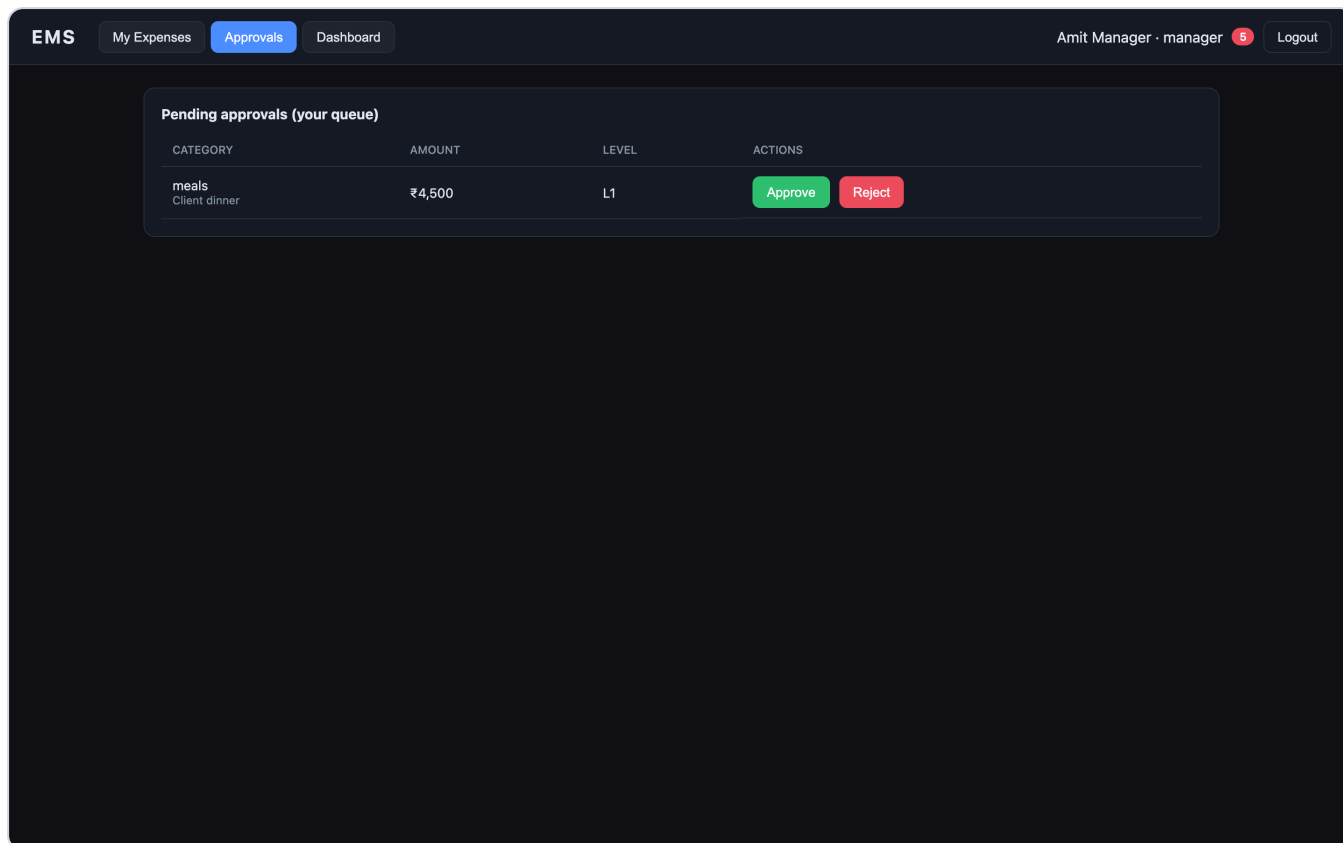
5. Approvals

GET /approvals/pending — my approval queue

Permission: expense:approve . Returns steps assigned to you that are pending at the expense's current level. **Response** 200

```
[
  {
    "id": "step-uuid",
    "expense_id": "uuid",
    "level": 1,
    "required_role": "manager",
    "status": "pending",
    "amount": "4500.00",
    "base_amount": "4500.00",
    "currency": "INR",
    "type": "reimbursement",
    "category": "meals",
    "description": "Client dinner",
    "requester_id": "uuid",
    "expense_status": "in_review"
  }
]
```

UI — Manager approvals queue: a manager sees only the steps assigned to them at the current level, and can Approve or Reject (reason prompted) directly.



6. Attachments

POST /attachments/presign — mock S3 presigned URL

Request: `{ "filename": "bill.png", "contentType": "image/png" }` **Response** `200`

```
{
  "key": "ems-attachments/<org>/<uuid>-bill.png",
  "uploadUrl": "https://mock-s3.local/...?X-Amz-Signature=mock&expires=900",
  "method": "PUT", "expiresInSeconds": 900,
  "note": "MOCK presigned URL (no real S3 in prototype)"
}
```

Errors: `400` missing filename/contentType.

7. Policies

GET /policies

Permission: `analytics:view` . **Response** `200` → array:

```
[{ "id":"uuid","name":"Default approval policy","tolerance_percent":"10.00","active":true,"version":
  "rules_json":{ "currency":"INR","rules":[
    {"min":0,"max":5000,"levels":["manager"]},
```

```
{"min":5001,"max":50000,"levels":["manager","finance"]},
{"min":50001,"max":null,"levels":["manager","finance","admin"]}]}}}]
```

POST /policies — create

Permission: `policy:manage` . Request

```
{ "name": "Travel policy", "tolerancePercent": 10,
  "rulesJson": { "currency":"INR", "rules":[ {"min":0,"max":10000,"levels":["manager"]} ] } }
```

Response `201` policy object. **Errors:** `400` invalid · `422` a rule's `max < min` .

PATCH /policies/:id — update (bumps version)

Any subset of `name` , `tolerancePercent` , `active` , `rulesJson` . `404` if missing.

DELETE /policies/:id

Response `200` → `{ "deleted": true }` . `404` if missing.

UI — Policies: the configurable approval rules per organization (amount range → approver chain) plus the tolerance percentage.

The screenshot shows the 'Policies' tab in the EMS application. The header includes 'EMS' and navigation links: 'My Expenses', 'Approvals', 'Dashboard', and 'Policies'. The user is 'Neha Finance' with a 'finance' role and 2 notifications. The main content area is titled 'Approval policies' and contains a table with the following data:

NAME	RULES (AMOUNT → LEVELS)	TOLERANCE	ACTIVE
Default approval policy	0-5000: manager 5001-50000: manager → finance 50001-∞: manager → finance → admin	10.00%	true

8. Budgets

GET /budgets (perm `analytics:view`) → array of budgets.

POST /budgets (perm budget:manage)

Request: { "userId": "uuid|null", "scope": "user|org", "period": "daily|monthly", "limitAmount": 50000, "currency": "INR" } **Response** 201 budget object. **Errors:** 400 invalid.

GET /budgets/utilization — current user's usage

Response 200 → { "monthly": { "limit": 2000000, "spent": 25200, "pct": 1 } }

9. Users

GET /users (perm user:manage) → array (password hashes stripped).

POST /users (perm user:manage)

Request: { "name":"New", "email":"new@acme.test", "password":"secret1", "role":"employee", "managerId":"uuid|null" } **Response** 201 user (no hash). **Errors:** 400 invalid · 409 duplicate email.

PATCH /users/:id (perm user:manage)

Any subset of name , role , managerId , isActive . 404 if missing.

UI — Admin Users: admins add employees/managers/finance to their org (with optional manager assignment) and view the org roster.

NAME	EMAIL	ROLE
Aditi Admin	admin@acme.test	admin
Neha Finance	cfo@acme.test	finance
Amit Manager	manager@acme.test	manager
Riya Sharma	riya@acme.test	employee
Arjun Verma	arjun@acme.test	employee

10. Notifications

GET /notifications → my notifications (newest 100).

```
[{ "id":"uuid","type":"approval_requested","payload_json":{"expenseId":"uuid","level":1},"read":false
```

POST /notifications/:id/read → { "read": true } . 404 if not yours.

11. Analytics / Observability

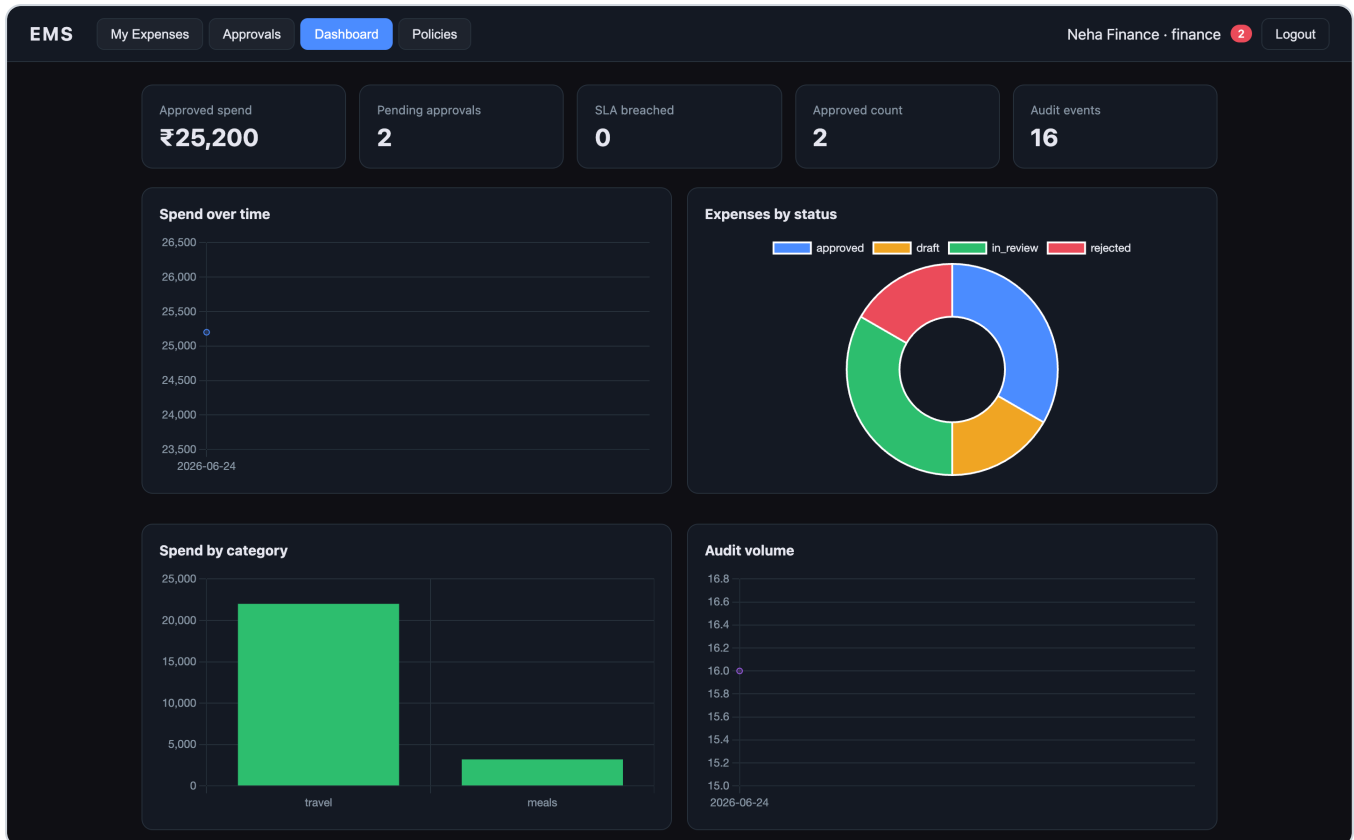
All require analytics:view . Summary is Redis/memory-cached (short TTL).

Endpoint	Response
GET /analytics/summary	{ approvedSpend, approvedCount, pendingApprovals, slaBreached, auditEvents, byStatus:{...} }
GET /analytics/spend	[{ "day":"2026-06-24", "total": 25200 }]
GET /analytics/by-status	[{ "status":"approved", "count": 2 }]
GET /analytics/by-category	[{ "category":"travel", "total": 22000 }]
GET /analytics/audit-volume	[{ "day":"2026-06-24", "count": 16 }]

Example GET /analytics/summary :

```
{ "approvedSpend": 25200, "approvedCount": 2, "pendingApprovals": 2, "slaBreached": 0, "auditEvents": 16, "byStatus": { "approved": 2, "draft": 1, "in_review": 2, "rejected": 1 } }
```

UI — Finance/Admin observability dashboard: stat cards (approved spend, pending, SLA breached, audit events) plus charts — spend over time, expenses by status, spend by category, audit volume. Data is aggregated from Postgres and cached.



12. Security, isolation & concurrency

Authentication. Access tokens are short-lived JWTs (`{ id, org_id, role, email }`). Refresh tokens are **opaque random strings stored only as SHA-256 hashes** in `refresh_tokens` . `/auth/refresh` does **single-use rotation** (old token revoked, new one issued, chained via `replaced_by`); replaying a revoked token is treated as theft and revokes the user's whole session family. `/auth/logout` revokes a token. (Code: `src/auth/refreshToken.ts` .)

Tenant isolation. `org_id` is read from the verified token (never the client) and applied to every query — the core isolation guardrail. This is app-layer isolation; one missing predicate would leak across tenants. The production hardening is **Postgres Row-Level Security**:

```
ALTER TABLE expense_requests ENABLE ROW LEVEL SECURITY;
CREATE POLICY tenant_isolation ON expense_requests
  USING (org_id = current_setting('app.org_id')::uuid);
-- set per transaction: SET LOCAL app.org_id = '<org from token>';
```

It's documented rather than enabled here because RLS interacts with the connection pool and the migration/seed paths; the app-layer guard plus this plan is the deliberate tradeoff.

Concurrency (no double-apply). Every state transition (submit/edit/approve/reject/withdraw) opens a transaction and takes an exclusive row lock via `SELECT ... FOR UPDATE ( getByIdForUpdate )` . Concurrent approvals on the same expense are serialized by Postgres: the loser blocks, re-reads the committed state, and is rejected by the status/level guards. Idempotency keys are orthogonal — they dedupe identical retries, the row lock handles distinct concurrent requests.

13. End-to-end flow (recap)

```
signup/login → create expense (draft) → submit
  → engine reads policy, snapshots it, builds ApprovalStep chain by amount
  → notify approver(s)
manager approve → (advance level) → finance approve → ... → APPROVED
                |
                reject (reason) → REJECTED
every transition → immutable audit_log entry (in the same DB transaction)
finance/admin → dashboard reads aggregates for live observability
```